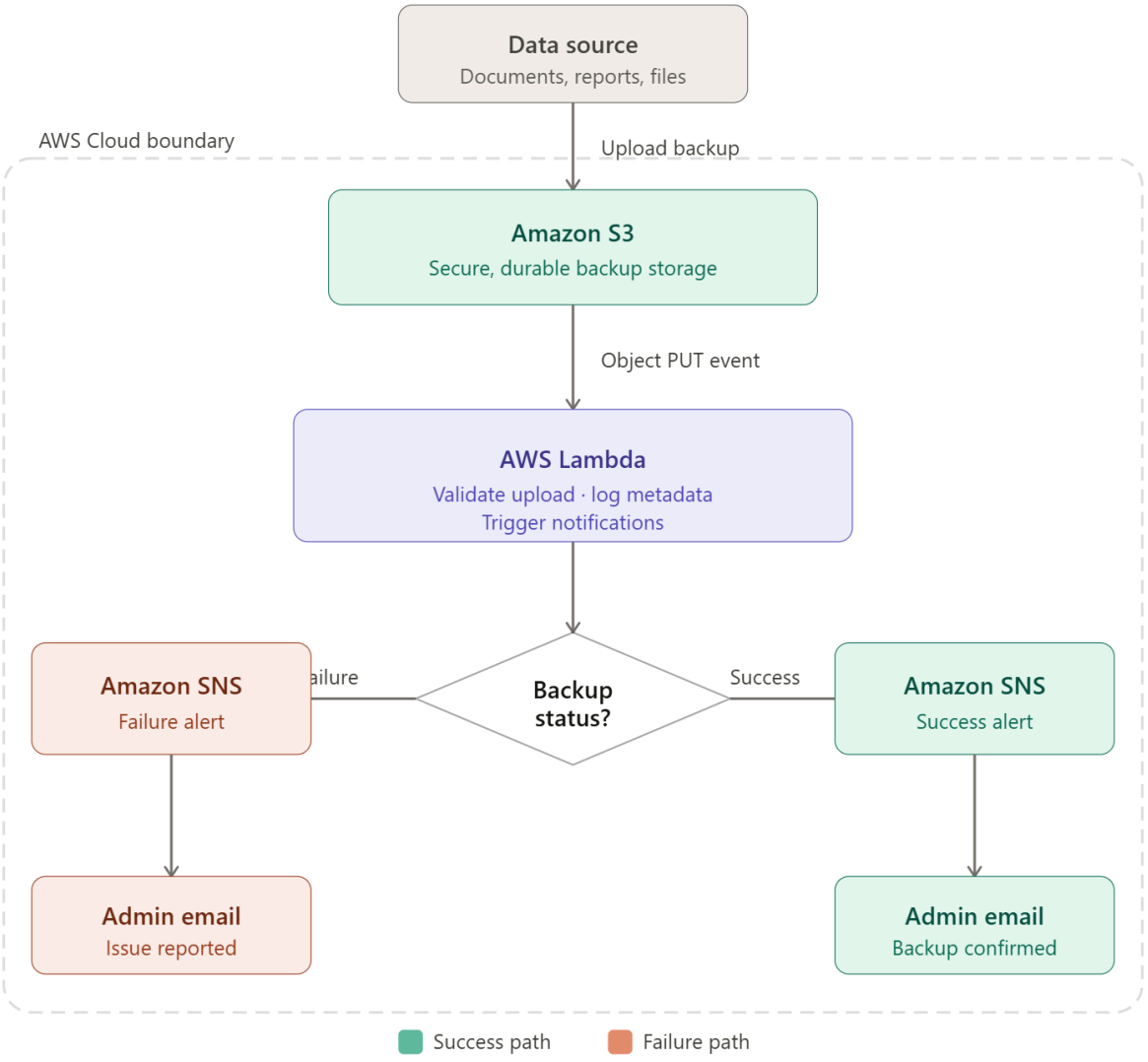
Automating Secure Data Backup and Real-Time Notifications Using Amazon S3, AWS Lambda, and Amazon SNS

Objective :

Design and implement a secure, automated cloud backup system using Amazon S3, AWS Lambda, and Amazon SNS to ensure reliable data storage, real-time monitoring, and instant notifications for backup status.



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1. Architecture Overview

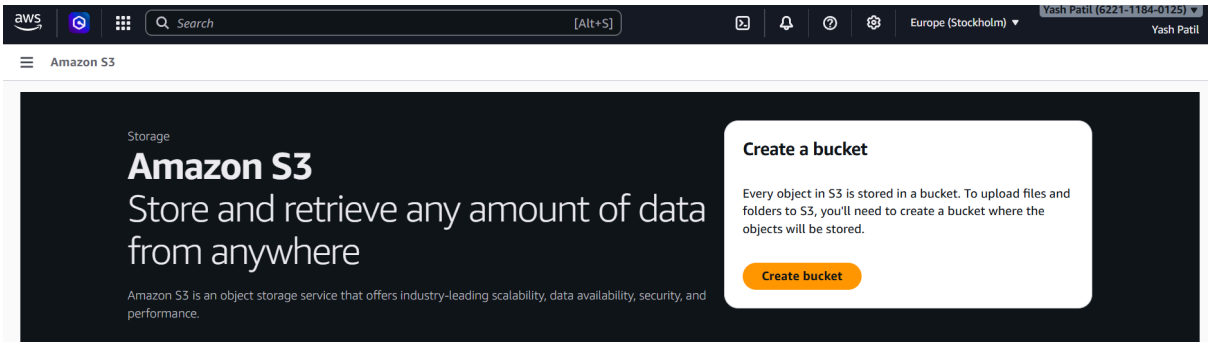
Workflow:

1. Files uploaded → **Amazon S3 bucket**
2. Event triggers → **AWS Lambda**
3. Lambda processes/validates backup
4. Notification sent via → **Amazon SNS**

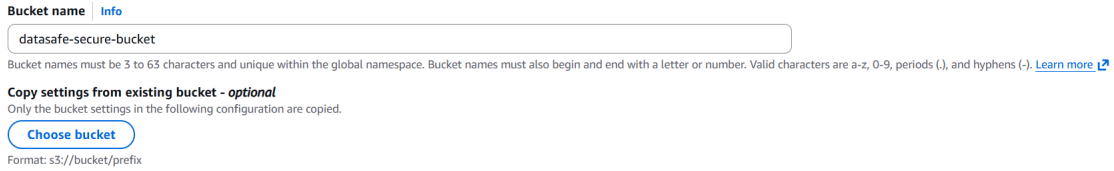
2. Step-by-Step Implementation

Step 1: Create S3 Bucket (Storage Layer)

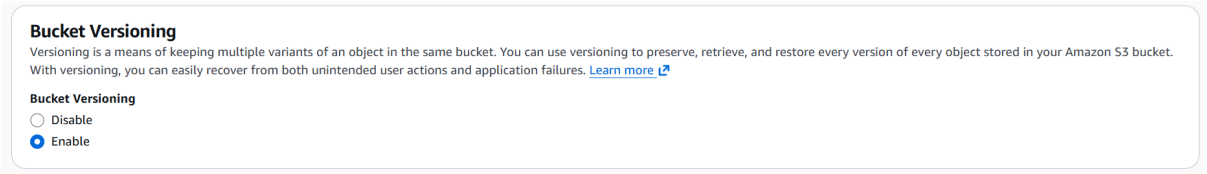
- Go to AWS Console → S3



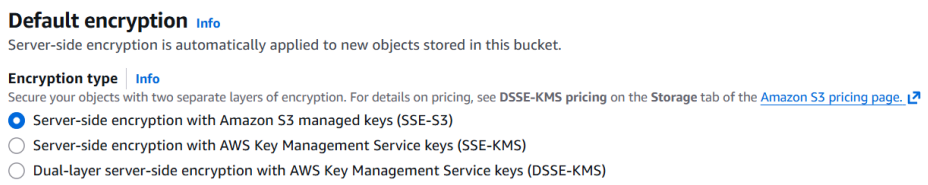
- Create bucket:
  - Unique name (`datasafe-secure-bucket`)



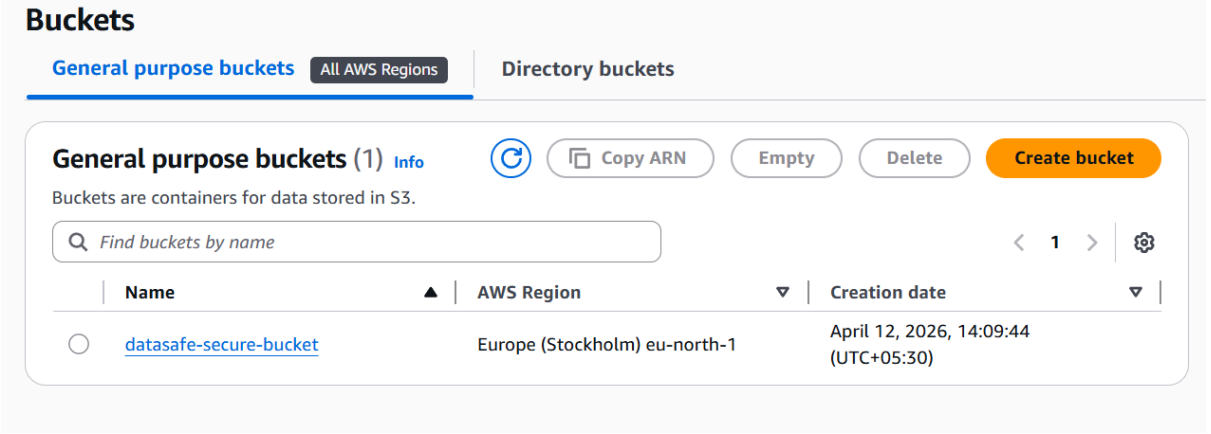
  - Enable versioning (important for recovery)



  - Enable encryption (SSE-S3 or SSE-KMS)

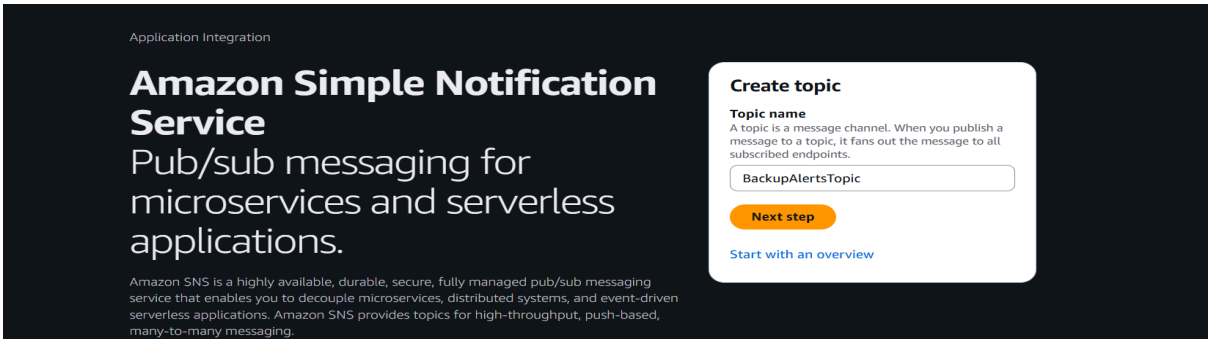


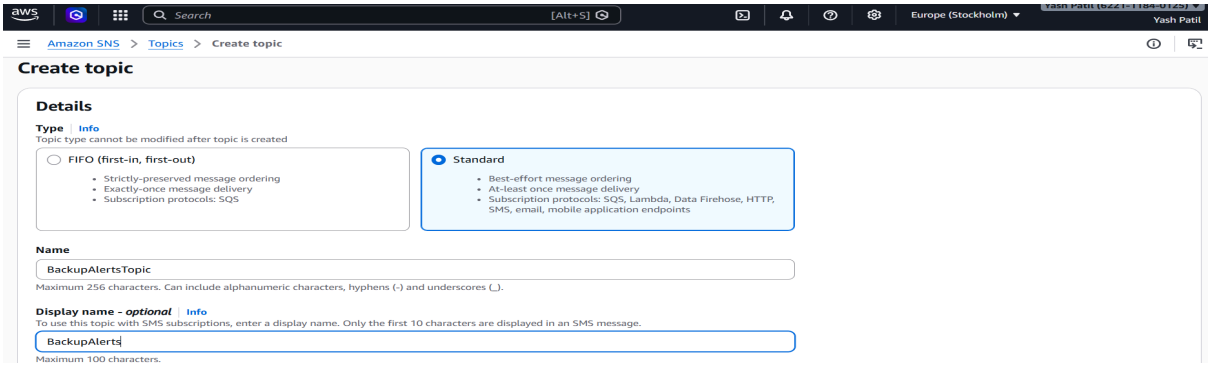
Secure, durable backup storage

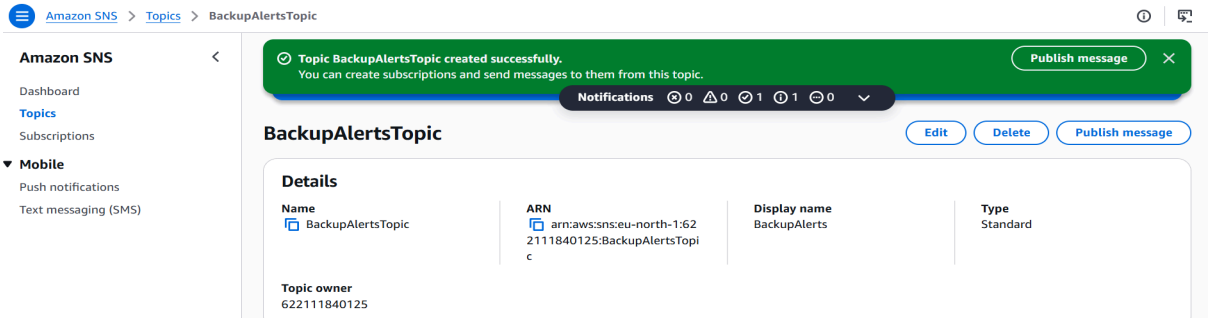


Step 2: Create SNS Topic (Notification System)

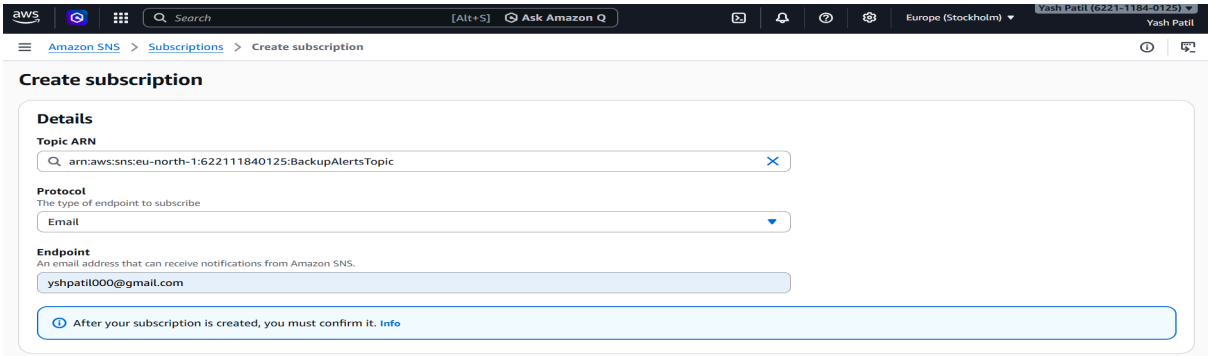
- Go to SNS → Create Topic
  - Type: Standard
  - Name: `BackupAlertsTopic`

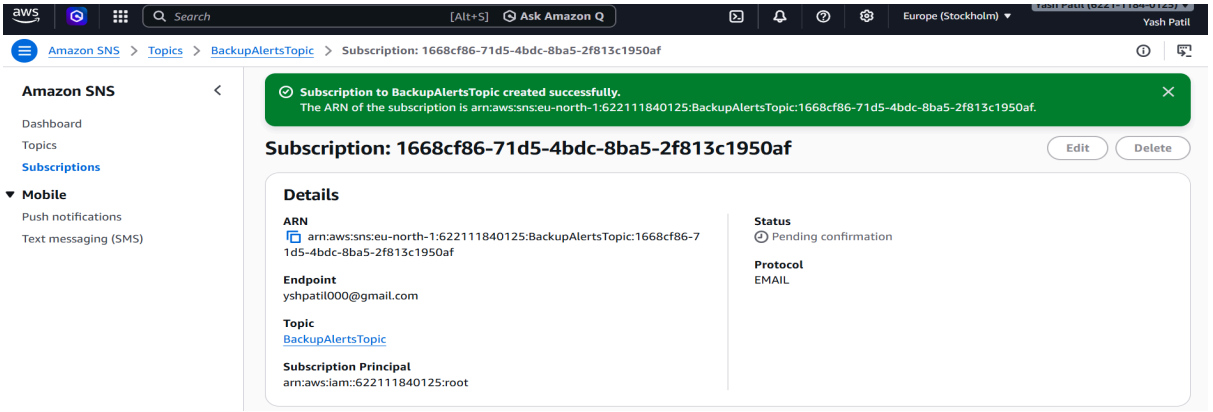




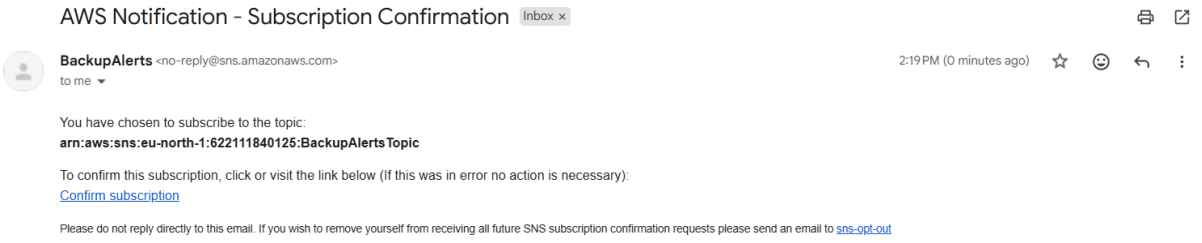


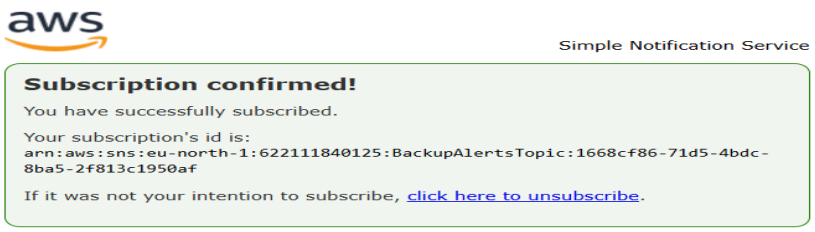
- Create Subscription:
  - Protocol: Email
  - Enter admin email → Confirm subscription

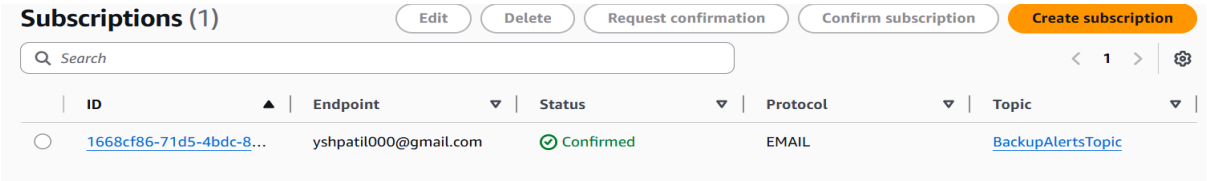




- Go to your **email inbox**
- You will receive a message from AWS
- Click **"Confirm subscription"**





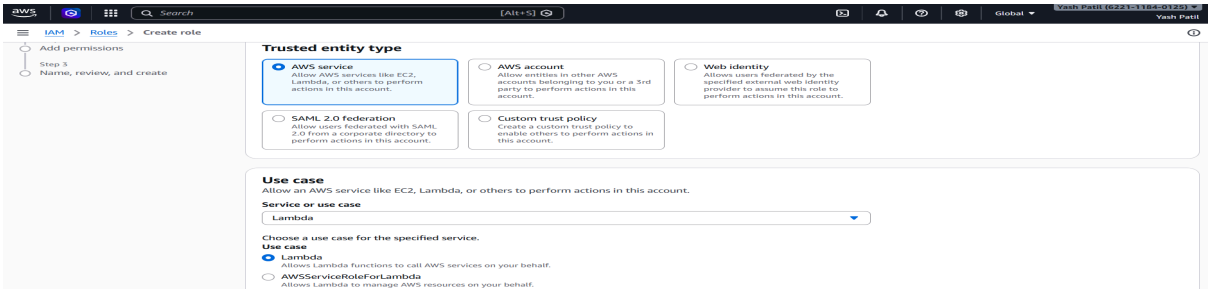


Purpose:

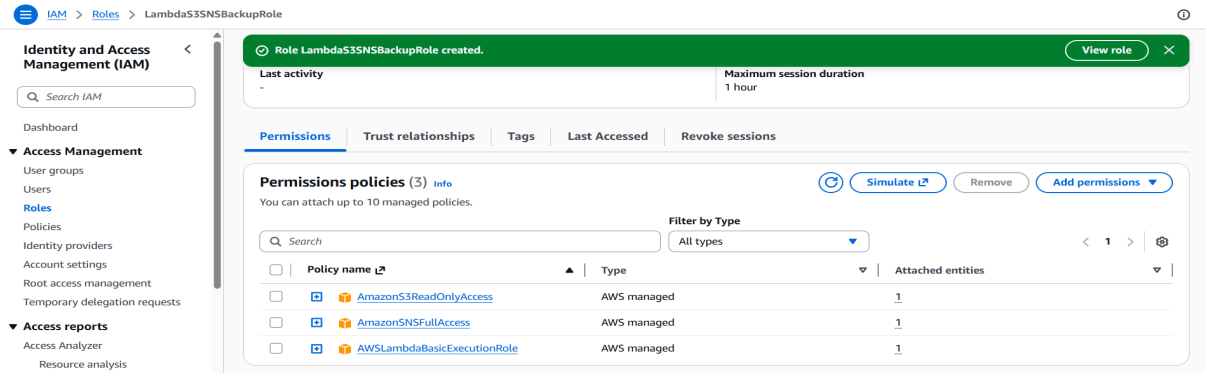
- Sends alerts for:
  - Backup success
  - **Backup failure/errors**

Step 3: Create IAM Role (Permissions)

- Go to IAM → Roles → Create Role



- Attach policies:
  - `AmazonS3ReadOnlyAccess`
  - `AmazonSNSFullAccess`
  - `AWSLambdaBasicExecutionRole`

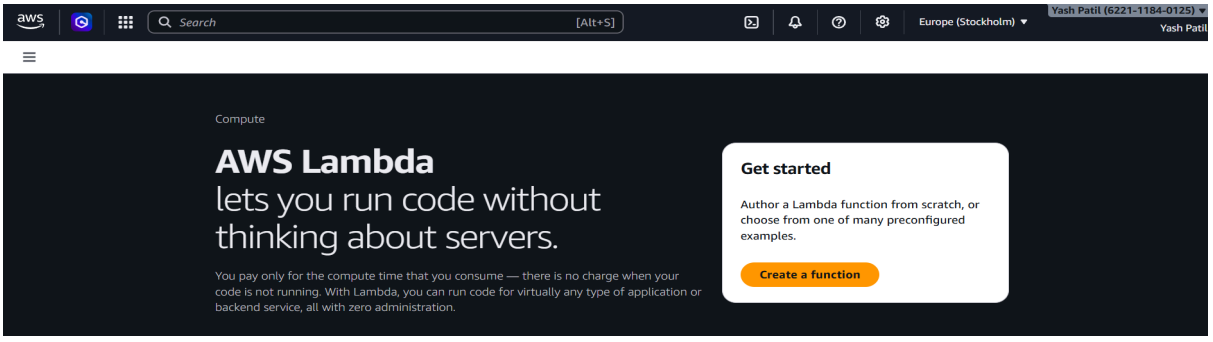


Purpose:

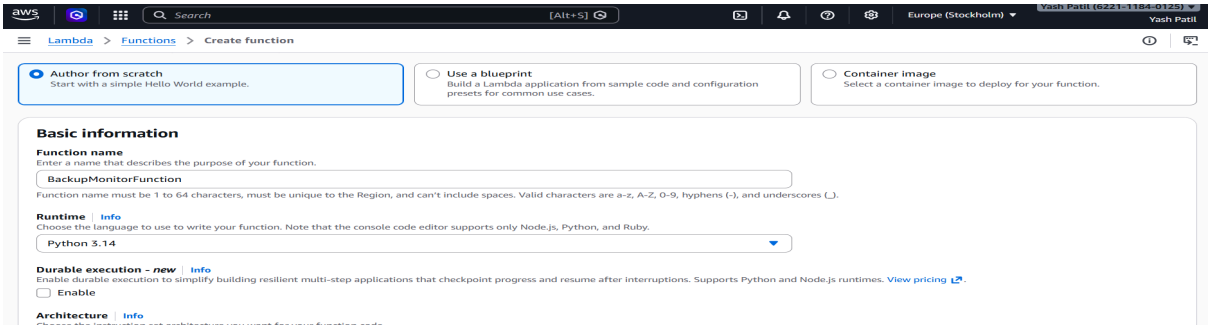
- Allows Lambda to:
  - Read S3 events/files
  - Publish notifications to SNS

Step 4: Create Lambda Function (Automation Logic)

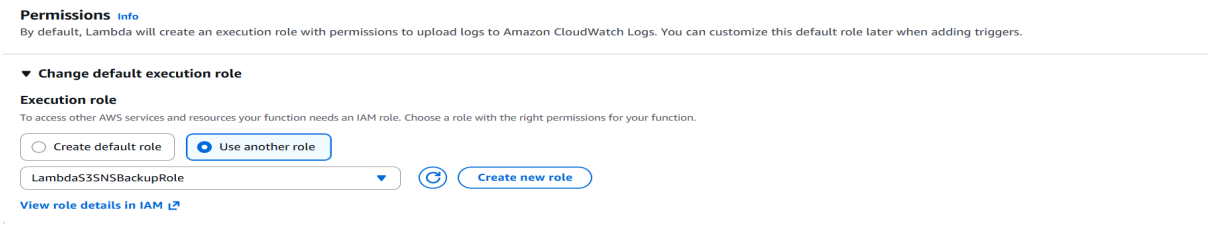
Go to Lambda → Create Function

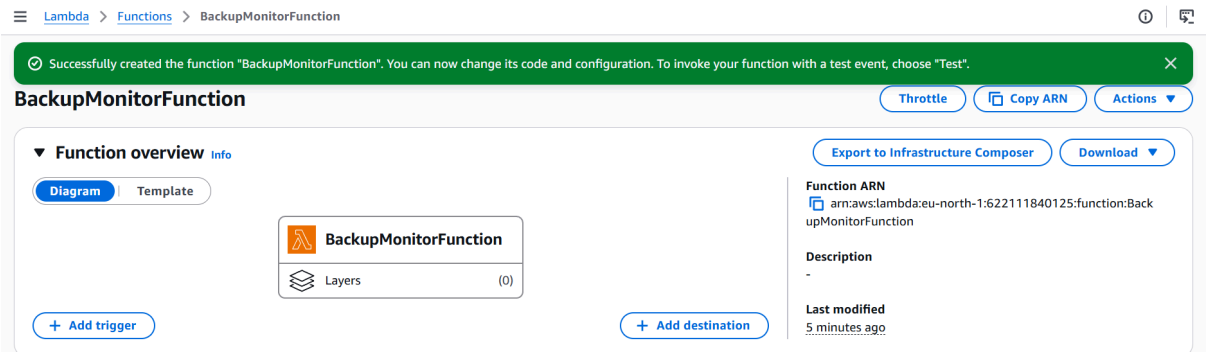


- Runtime: Python / Node.js
- Name: `BackupMonitorFunction`



- Attach IAM role

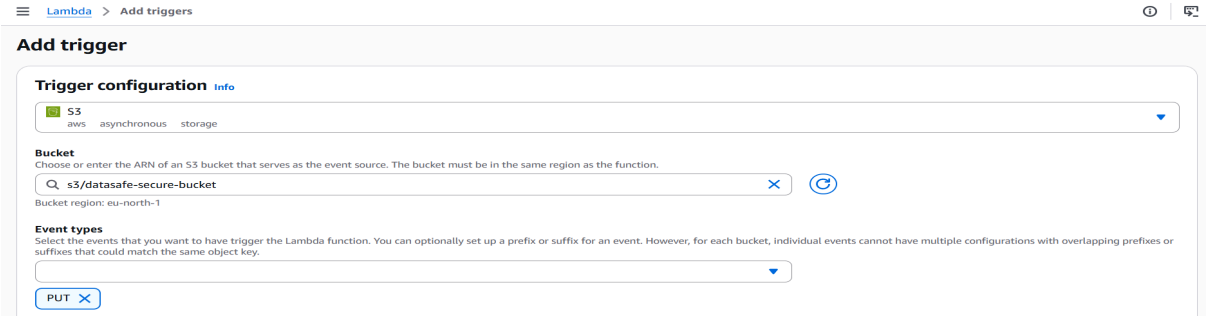


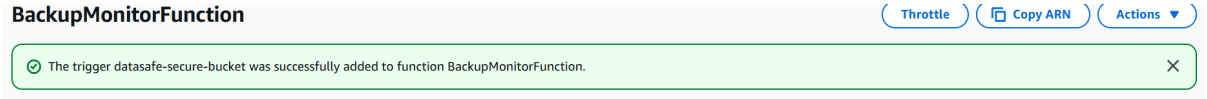


Step 5: Add S3 Trigger to Lambda

In Lambda → Add Trigger

- Source: S3
- Event Type: PUT (file upload)
- Select bucket





Step 6: Lambda Function Code (Python)

```python
import json
import boto3

sns = boto3.client('sns')

SNS_TOPIC_ARN = "arn:aws:sns:eu-north-1:622111840125:BackupAlertsTopic"

def lambda_handler(event, context):
    try:
        # Extract file details
        bucket = event['Records'][0]['s3']['bucket']['name']
        file_name = event['Records'][0]['s3']['object']['key']

        message = f"Backup Successful!\nBucket: {bucket}\nFile: {file_name}"

        sns.publish(
            TopicArn=SNS_TOPIC_ARN,
            Message=message,
            Subject="Backup Success Notification"
        )
```

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Deploy the Code

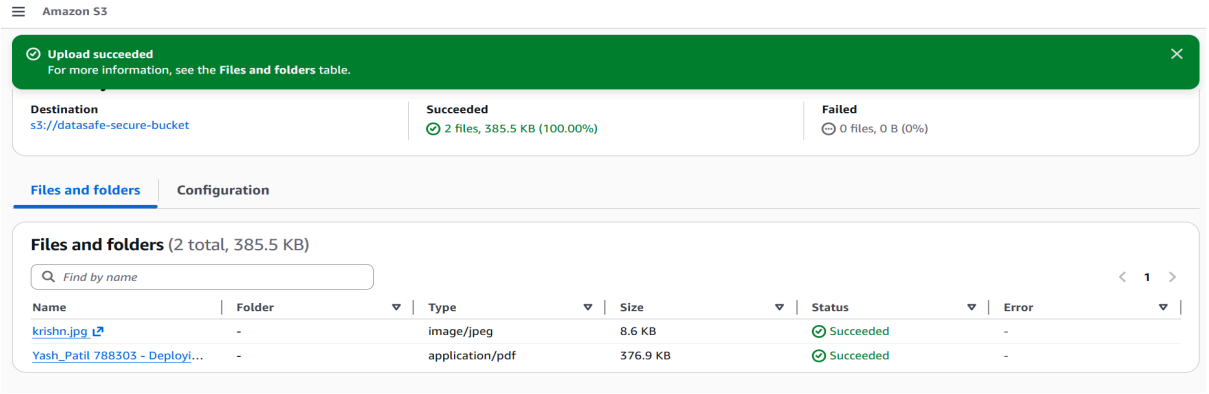
- Click Deploy in AWS Lambda
- This saves and activates your function

Verification :

1. Upload Test File

In **Amazon S3**:

- Open bucket
- Upload file

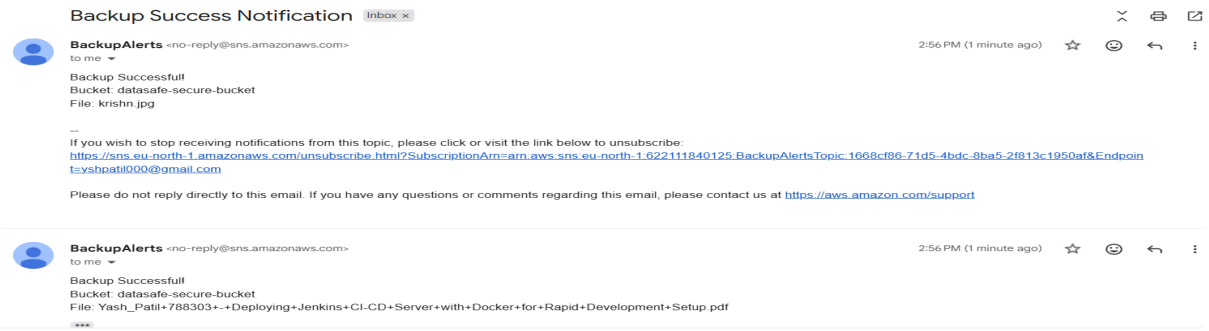


2. Verify Email Notification

Check email inbox

You should receive:

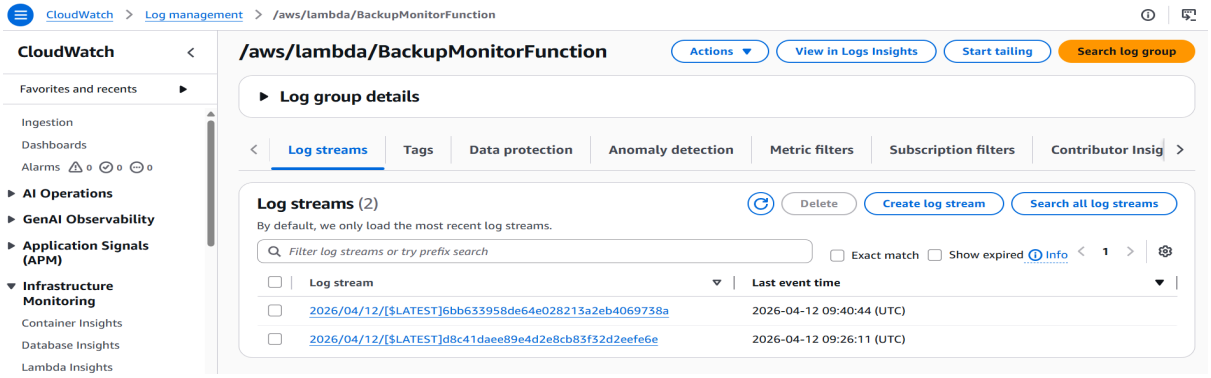
- **Subject:** Backup Success Notification
- Contains bucket name & file name



3. Verify Lambda Execution Logs

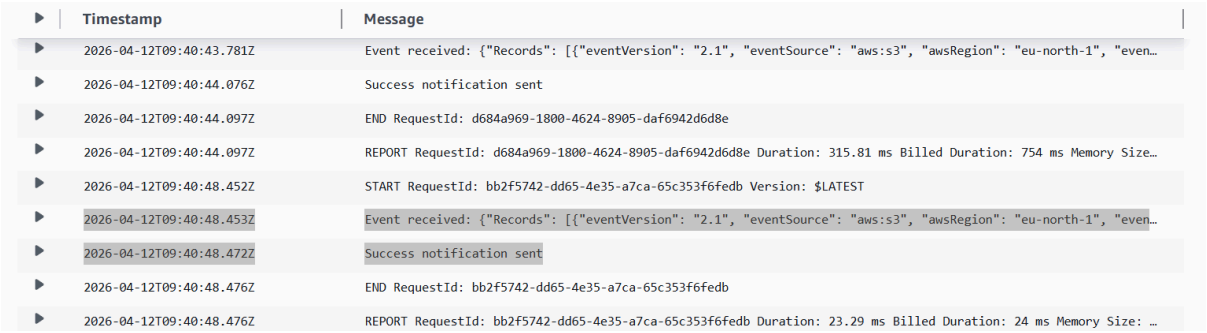
Go to:

- Lambda → **Monitor** → **CloudWatch Logs**



Check:

- Logs show:
  - `Event received`
  - `Success notification sent`



Timestamp	Message
2026-04-12T09:40:43.781Z	Event received: {"Records": [{"eventVersion": "2.1", "eventSource": "aws:s3", "awsRegion": "eu-north-1", "even…
2026-04-12T09:40:44.076Z	Success notification sent
2026-04-12T09:40:44.097Z	END RequestId: d684a969-1800-4624-8905-daf6942d6d8e
2026-04-12T09:40:44.097Z	REPORT RequestId: d684a969-1800-4624-8905-daf6942d6d8e Duration: 315.81 ms Billed Duration: 754 ms Memory Size…
2026-04-12T09:40:48.452Z	START RequestId: bb2f5742-dd65-4e35-a7ca-65c353f6fedb Version: $LATEST
2026-04-12T09:40:48.453Z	Event received: {"Records": [{"eventVersion": "2.1", "eventSource": "aws:s3", "awsRegion": "eu-north-1", "even…
2026-04-12T09:40:48.472Z	Success notification sent
2026-04-12T09:40:48.476Z	END RequestId: bb2f5742-dd65-4e35-a7ca-65c353f6fedb
2026-04-12T09:40:48.476Z	REPORT RequestId: bb2f5742-dd65-4e35-a7ca-65c353f6fedb Duration: 23.29 ms Billed Duration: 24 ms Memory Size: …

Conclusion

The project successfully implements a secure and automated cloud-based backup monitoring system using Amazon S3, AWS Lambda, and Amazon SNS.

All backup files are reliably stored in S3, while Lambda ensures real-time processing and validation whenever new data is uploaded. The integration with SNS enables instant email notifications, keeping administrators informed about backup status without manual intervention.

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